增益 NLSE

$$i\psi_z = \frac{\beta(z)}{2}\psi_{\tau\tau} - \gamma(z)|\psi|^2\psi + i\frac{g(z)}{2}\psi. \qquad (11)$$

引入无量纲函数 $u(s,t)$.

$$\psi(z,t) = \frac{|\rho(z)|^{1/2}}{\tau_0[1-c_0 D(z)]}\exp[i\Phi(z,\tau)]u(s,t)$$

$$\therefore \quad \Phi(z,\tau) = a_0 - \frac{\omega^2}{4}D(z) - \omega[\tau-\tau_c(z)] + \frac{c_0[\tau-\tau_c(z)]^2}{1-c_0 D(z)}.$$

$$t(z,\tau) = \frac{\tau-\tau_c(z)}{\tau_0(1-c_0 D(z))}, \quad s(z) = \frac{|D(z)|}{2\tau_0^2(1-c_0 D(z))}$$

$$\rho(z) = \frac{\beta(z)}{\gamma(z)}, \qquad D(z) = 2\int_0^z \beta(z')\,dz'$$

$$\tau_c = \tau_c^0 + \frac{\omega}{2}D(z).$$

· ω 给出了一般情况下的行进波解:

$$\tilde{\psi}(z,\tau) = \psi\left(z, \tau-\frac{\omega}{2}D(z)\right)\exp\left[i\left(\frac{\omega^2}{4}D(z)-\omega\tau\right)\right] \qquad (16)$$

· $\psi(z,\tau)$ 为 NLSE 的任意解.

· $\tilde{\psi}(z,\tau)$ 是相对 $\psi(z,\tau)$ 的平移解.

特殊情形下, $\beta(z)=\beta=$ const. 变换 (16) 简化为:

$$\tilde{\psi}^{(s)}(z,\tau) = \psi^{(s)}(z,\tau-vz)\exp\left[i\left(\frac{v^2}{2\beta}z - \frac{v}{\beta}\tau\right)\right]$$